

ORCS 4529: Reinforcement Learning

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Dynamic Programming (DP) based algorithms for RL

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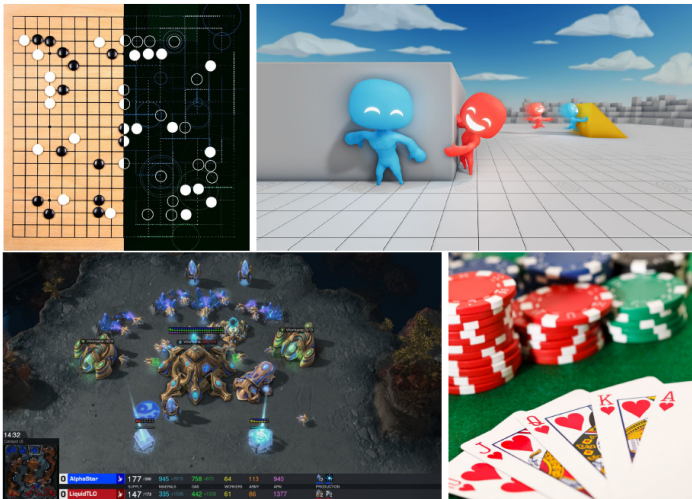
Multi-Agent Reinforcement Learning (MARL)

Game Theory (a fast track introduction)

Markov Games

RL with multiple (competing) agents

The environment is no longer external – responds and adapts to the RL agent.



Classic Game Theory: One Shot Games

Two or more players make one decision each.

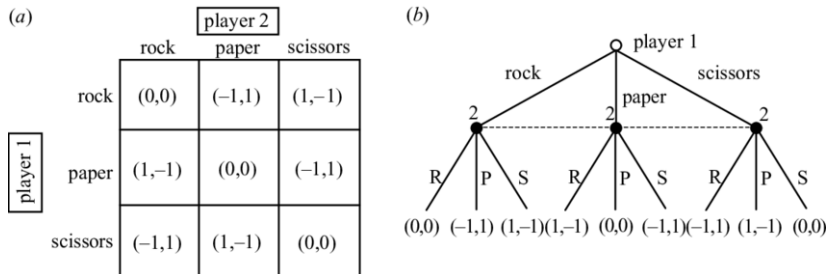
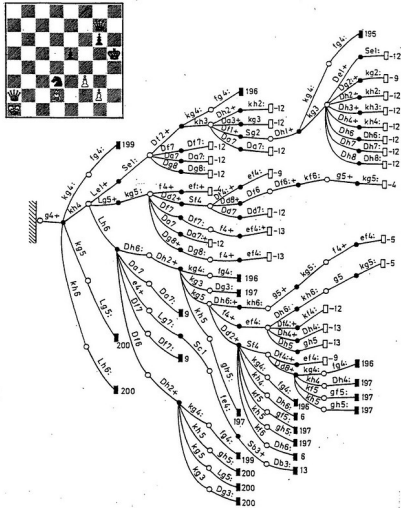


Figure: (a) Normal form (b) extensive form

Equilibrium solution concepts, and algorithms for computing equilibrium strategies

Multi-stage games



Multi-agent RL = Games + Sequential Decision Making

Games

A game is defined by:

- ▶ A set of n players P ,
- ▶ set of actions A_i for each player i , $A = A_1 \times A_2 \times \cdots \times A_n$, and
- ▶ a utility function $u_i : A \rightarrow \mathbb{R}$ for each player i .

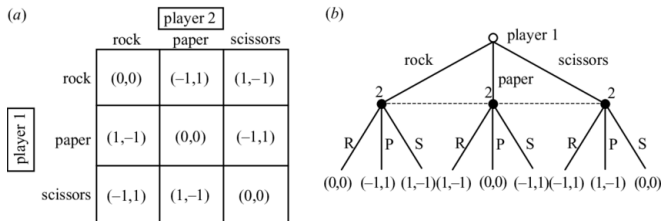


Figure: Rock-Paper-Scissors (a) Normal form (Payoff Matrix) representation (b) Extensive form

How would a rational player behave?

- ▶ Best response of player i to actions $a_{-i} \in A_{-i}$ of other players:

$$a_i \in \arg \max_{a \in A_i} u_i(a, a_{-i}).$$

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If an action is a dominant action, it is a best response in all situations! It never hurts to play this action. Such action may not exist though.

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$$a_i \in \arg \max_{a \in A_i} u_i(a, a_{-i}).$$

- ▶ An action a for player i (weakly) dominates another action a' if for all $a_{-i} \in A_{-i}$,

$$u_i(a, a_{-i}) \geq u_i(a', a_{-i})$$

with strict inequality for some $a_{-i} \in A_{-i}$.

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- ▶ If an action is weakly dominated by another action, it should never be played.
- ▶ An action is dominant for player i if it weakly dominates all other actions. (best response in all situations! may not exist though)

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► If an action is weakly dominated by another action, it should never be played.

► An action is dominant for player i if it weakly dominates all other actions. (best response in all situations! may not exist though)

If an action is a dominant action, it is a best response in all situations! It never hurts to play this action. Such action may not exist though.

Examples

	Confess	Silent
Confess	$(1, 1)$	$(5, 0)$
Silent	$(0, 5)$	$(3, 3)$

Figure: Prisoner's Dilemma

_____ is a dominant strategy for each player.

	Football	Opera
Football	$(5, 1)$	$(0, 0)$
Opera	$(0, 0)$	$(1, 5)$

Figure: Battle of the Sexes

Examples

	Confess	Silent
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"Confess" is a dominant strategy for each player.

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Figure: Battle of the Sexes

No dominant Strategy.

Solution Concepts

General idea: For a game to be in a stable situation (aka equilibrium), all players should be playing a best response. (Otherwise, by definition, the situation would not be stable since somebody would want to change their action.)

- ▶ Dominant Strategy Equilibrium
- ▶ Nash Equilibrium
- ▶ Correlated Equilibrium
- ▶ Coarse Correlated Equilibrium

Dominant Strategy Equilibrium

A profile of actions $a = (a_1, \dots, a_n)$ is a dominant strategy equilibrium if a_i is a dominant strategy for every player i .

	Confess	Silent
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	Football	Opera
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Opera	(0, 0)	(1, 5)

	STOP	GO
STOP	(0, 0)	(0, 1)
GO	(1, 0)	(-100, -100)

Figure: Battle of the Sexes, Traffic Light

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No dominant strategy equilibrium

Dominant Strategy Equilibrium

	rock	paper	scissors
rock	(0,0)	(-1,1)	(1,-1)
paper	(1,-1)	(0,0)	(-1,1)
scissors	(-1,1)	(1,-1)	(0,0)

Figure: Rock Paper Scissors

	Heads	Tails
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Figure: Matching Pennies

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No dominant Strategy equilibrium

Nash Equilibrium

Definition

A profile of actions $a = (a_1, \dots, a_n)$ is a
(pure strategy) Nash Equilibrium if every player i

$$u_i(a_i, a_{-i}) \in \arg \max_{a'_i \in A_i} u_i(a'_i, a_{-i})$$

i.e. simultaneously, all players are playing a best response to one another.

Examples

	Confess	Silent
Confess	(1, 1)	(5, 0)
Silent	(0, 5)	(3, 3)

Figure: Prisoner's Dilemma

	Football	Opera
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For the Prisoner's dilemma, "(Confess, Confess)" is a dominant strategy equilibrium and a pure strategy Nash Equilibrium (PSNE).

For Battle of the sexes and Traffic Light, No dominant strategy equilibrium but pure strategy Nash equilibrium exists. Battle of the sexes has two pure strategy NE: Football, Football, and Opera, Opera. Traffic light has two pure strategy NE: stop go and go stop.

Examples

	rock	paper	scissors
rock	$(0,0)$	$(-1,1)$	$(1,-1)$
paper	$(1,-1)$	$(0,0)$	$(-1,1)$
scissors	$(-1,1)$	$(1,-1)$	$(0,0)$

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No dominant strategy equilibrium and No Pure strategy Nash equilibrium, Mixed Strategy equilibrium exists. In RPS, it is $1/3, 1/3, 1/3$. For matching pennies it is $1/2, 1/2$.

For others, each DSE and PSNE is a MSNE, but there are additional MSNE, which may be better or worse in certain ways. For example for traffic light has PSNE (*STOP*, *GO*) and (*GO*, *STOP*) but also a MSNE where each player stops with probability $100/101$ and goes with probability $1/101$. This has expected utility 0 for both players. But it is an equilibrium because any player unilaterally deviating from this strategy profile will get negative utility.

Mixed Strategy Nash Equilibrium

- ▶ A mixed-strategy $p_i \in \Delta^{A_i}$ is a probability distribution over actions of player i .

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- ▶ Define the utility of a mixed strategy profile for player i as the expected utility

$$u_i(p_1, \dots, p_n) = \mathbb{E}_{a_1 \sim p_1, \dots, a_n \sim p_n} [u_i(a_1, \dots, a_n)]$$

where $a_1, \dots, a_n$ are sampled **independently** from $p_1, \dots, p_n$.

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where $a_1, \dots, a_n$ are sampled **independently** from $p_1, \dots, p_n$.

- ▶ A mixed strategy Nash equilibrium is a tuple of mixed strategies $p = (p_1, \dots, p_n)$ such that for all i

$$u_i(p_i, p_{-i}) \geq u_i(a_i, p_{-i})$$

That is, each player i 's mixed strategy is a best response to the mixed strategy of other players.

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Seminal result by Nash

Theorem

Every game with a finite set of players and actions has a mixed strategy Nash equilibrium.

Examples

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Correlated equilibrium

	STOP	GO
STOP	(0,0)	(0,1)
GO	(1,0)	(-100,-100)

Has two dominant strategy equilibrium and a mixed strategy Nash Equilibrium but none of them are satisfactory solutions to this problem.

What if the two players could behave in a more coordinated manner?

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What if the two players could behave in a more coordinated manner?

This game has two pure strategy Nash equilibria: $(GO, STOP)$, and $(STOP, GO)$ but these are clearly not ideal because there is one player who never gets any utility. There is also a mixed strategy Nash equilibrium: which is even worse since both get 0 utility. Ideally, the action profile we want is where $(STOP, GO)$ happens with 50% probability and $(GO, STOP)$ happens with 50% probability. Then, each player gets $1/2$ utility. The problem is that no mixed strategy would create such profiles – since we require the two players to sample independently.

Correlated equilibrium

No longer require that each player actions are sampled independently from other players.

Definition

A correlated equilibrium is a (joint) distribution D over action profiles $A = A_1 \times \cdots \times A_n$ such that for every player i ,

$$\mathbb{E}_{a \sim D}[u_i(a_i, a_{-i})] \geq \mathbb{E}_{a \sim D}[\max_{a'_i} \mathbb{E}[u_i(a'_i, a_{-i}) | a_i]]$$

That is, no player can benefit by uniformly swapping their action a_i with another action a'_i in every situation where they were asked to play a given action a_i .

Correlated equilibrium

	STOP	GO
STOP	(0,0)	(0,1)
GO	(1,0)	(-100,-100)

Correlated equilibrium: $(STOP, GO)$ and $(GO, STOP)$ each with probability $1/2$.

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Correlated equilibrium: (STOP, GO) and (GO, STOP) each with probability 1/2.

Intuitively, this is the equilibrium implemented by humans. Whenever player 1 is asked to play STOP, the other player is asked to play GO, so the player knows it would not benefit by playing GO in those situations.

Coarse Correlated equilibrium

Definition

A coarse correlated equilibrium is a (joint) distribution D over action profiles $A = A_1 \times \cdots \times A_n$ such that for every player i ,

$$\mathbb{E}_{a \sim D}[u_i(a_i, a_{-i})] \geq \max_{a'_i} \mathbb{E}_{a \sim D}[u_i(a'_i, a_{-i})]$$

Further relaxes to CE condition to allow action profile where no player can benefit *in expectation* by always playing some other action a'_i .

Example

	rock	paper	scissors
rock	(0,0)	(-1,1)	(1,-1)
paper	(1,-1)	(0,0)	(-1,1)
scissors	(-1,1)	(1,-1)	(0,0)

Figure: Rock paper scissors

Consider distribution D where all non-diagonal profiles are played with equal probability $1/6$. This is CCE but not CE.

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Example

	rock	paper	scissors
rock	(0,0)	(-1,1)	(1,-1)
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Figure: Rock paper scissors

Consider distribution D where all non-diagonal profiles are played with equal probability $1/6$. This is CCE but not CE.

When a player is asked to play P , they know the other player is playing R , S with equal probability, so they might want to change to R when ever asked to play P . However, since the opposition plays R, P, S with equal probability, 0 is the maximum utility any single action can achieve in expectation, which is same as each player's utility in this case.

Some notes

- ▶ $DSE \subset PSNE \subset MSNE \subset CE \subset CCE$
- ▶ Computational difficulty (when exists)

$$DSE \leq PSNE \leq MSNE \leq CE \leq CCE$$

- ▶ MSNE and hence CE, CCE always exist for finite games.
- ▶ Proof of existence of MSNE is non-constructive. How do we compute the desired equilibrium solution?

Zero Sum game and minimax theorem

A two player zero sum game is any two-player game such that

$$u_1(a_1, a_2) = -u_2(a_1, a_2), \forall a_1 \in A_1, a_2 \in A_2.$$

	rock	paper	scissors
rock	(0,0)	(-1,1)	(1,-1)
paper	(1,-1)	(0,0)	(-1,1)
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Figure: Rock paper scissors is a zero sum game

Two player zero sum game

- ▶ For a two player zero sum game, the game can be specified by a single utility function $u(a_1, a_2)$ which denotes say the utility of player 1, then player 2's utility is $-u(a_1, a_2)$.

Two player zero sum game

- ▶ For a two player zero sum game, the game can be specified by a single utility function $u(a_1, a_2)$ which denotes say the utility of player 1, then player 2's utility is $-u(a_1, a_2)$.
- ▶ For finite games, the function $u(a_1, a_2)$ can be represented as an $A_1 \times A_2$ matrix U , with $U_{ij} = u(i, j)$.

Two player zero sum game

- ▶ For a two player zero sum game, the game can be specified by a single utility function $u(a_1, a_2)$ which denotes say the utility of player 1, then player 2's utility is $-u(a_1, a_2)$.
- ▶ For finite games, the function $u(a_1, a_2)$ can be represented as an $A_1 \times A_2$ matrix U , with $U_{ij} = u(i, j)$.
- ▶ for continuous action spaces $x \in \mathcal{X}, y \in \mathcal{Y}$, utility $u(x, y)$.

Minmax Theorem (Von Neumann, 1928)

finite action spaces

For any finite two player zero sum game (matrix) U ,

$$\min_q \max_p p^T U q = \max_p \min_q p^T U q =: v^*$$

where v^* is referred to as the value of the game.

- ▶ Let p^*, q^* be a minmax solution. Then p^*, q^* is a (mixed strategy) Nash equilibrium
- ▶ Any Nash equilibrium is a minmax solution
- ▶ Utility of each player at all Nash equilibria is same and is called value of the game.

ORCS 4529: Reinforcement Learning

- └ Multi-Agent Reinforcement Learning (MARL)
 - └ Game Theory (a fast track introduction)
 - └ Minmax Theorem (Von Neumann, 1928)

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1. $(p^*)^T U q^* = \min_q \max_p p^T U q = \max_p p^T U q^* \geq p^T U q^*,$
 $(p^*)^T U q^* = \max_p \min_q p^T U q = \min_q (p^*)^T U q \leq (p^*)^T U q$
2. otherwise the max (min) player can go to the minmax solution and increase (decrease) their value

Note

Intuitively in zero sum game at Nash equilibrium both players are achieving the best possible utility they can hope for – assuming the opponent is smart.

May not hold for non zero-sum games:

	Football	Opera
Football	(5, 1)	(0, 0)
Opera	(0, 0)	(1, 5)

Figure: Two Nash equilibria: (5,1) and (1,5). In each case, only one player is achieving their best possible utility.

Minmax theorem for continuous action spaces

Given a function $u(x, y)$ that is continuous and convex in x and concave in y , we have

$$\min_{x \in A_1} \max_{y \in A_2} u(x, y) = \max_{y \in A_2} \min_{x \in A_1} u(x, y) =: v^*$$

where v^* is referred to as the value of the game.

- ▶ Any minmax solution x^*, y^* to the above problem is a (pure strategy) Nash equilibrium.

ORCS 4529: Reinforcement Learning

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Any Nash equilibrium is a minmax solution because otherwise the max (min) player can go to the minmax solution and increase (decrease) their value

Markov Games (aka Stochastic Games)

A Markov Game is defined by tuple $(\mathcal{S}, A = A_1 \times \cdots \times A_m, P, R)$
[Shapley 1953]

- ▶ $\mathcal{S}$: set of states
- ▶ A_i : set of actions for player i
- ▶ $P(s'|s, \mathbf{a})$: transition probability under action profile
 $\mathbf{a} = (a_1, \dots, a_n)$
- ▶ $R_i(s, \mathbf{a}) \in (0, 1)$, the expected reward for player i .

Policies

We assume all the past states and actions of both players are fully observable to both players.

Then, as in MDPs we have

- ▶ History-dependent policies
- ▶ Markovian policies (possibly randomized):

$$\pi_i : \mathcal{S} \rightarrow \Delta^{A_i}$$

- ▶ Non-stationary policies $\pi_i = \{\pi_{i,t}, t = 1, 2, 3, \dots\}$, where each $\pi_{i,t} : \mathcal{S} \rightarrow \Delta^{A_i}$.

Finite horizon H

Nash Equilibrium

Let $V_{i,h}^{\pi}(s)$ denote the expected reward for player i starting from state s at time index h under policy profile

$\pi = (\pi_{t,i}, \pi_{t,-i}, t = 1, \dots, H)$.

$$V_{i,h}^{\pi}(s) = \mathbb{E}\left[\sum_{t=h}^H R_i(s_t, \mathbf{a}_t) | s_h = s; \mathbf{a}_t = \pi_t(s_t)\right]$$

A policy profile π is in Nash equilibrium if for every player i , and all s, h ,

$$V_{i,h}^{\pi_i, \pi_{-i}}(s) \geq V_{i,h}^{\pi'_i, \pi_{-i}}(s), \quad \forall \pi'_i$$

ORCS 4529: Reinforcement Learning

- Multi-Agent Reinforcement Learning (MARL)
 - Markov Games
 - Finite horizon H

Finite horizon H

Nash Equilibrium

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$$V_{i,h}^{\pi}(s) = \mathbb{E} \left[\sum_{t=h}^H R_t(s_t, \mathbf{a}_t) \mid s_0 = s, \mathbf{a}_t = \pi_t(s_t) \right]$$

A policy profile π is in Nash equilibrium if for every player i , and all s, h ,

$$V_{i,h}^{\pi_i, \pi_{-i}}(s) \geq V_{i,h}^{\pi'_i, \pi_{-i}}(s), \quad \forall \pi'_i$$

Here the expectation is over $a_{i,t} \sim \pi_{i,t}$, $s_{t+1} \sim P(\cdot | s_t, \mathbf{a}_t)$. Actions are sampled independently for the m players from their respective policies.

Finite horizon

Nash Equilibrium by Dynamic Programming

- ▶ Initialize $V_{i,H+1}(s) = 0$ for all s .
- ▶ For $h = H - 1, \dots, 1$
 - ▶ Let $Q_{i,h}(s, \mathbf{a}) = R_i(s, \mathbf{a}) + \mathbb{E}_{s' \sim P(s, \mathbf{a})}[V_{i,h+1}(s')]$
 - ▶ For every s , consider the m player one-shot game with utility matrix $Q_{i,h}(s, \cdot)$: the utility of player i under action profile $\mathbf{a} = (a_i, a_{-i})$ is $Q_{i,h}(s, \mathbf{a})$.
 - ▶ Let $p^* = (p_1^*, \dots, p_m^*)$ be the Nash equilibrium of this one shot game. Define

$$\pi_{i,h}^*(s) := p_i^*$$

- ▶ Set $V_{i,h}(s) = \mathbb{E}_{\mathbf{a} \sim p^*}[Q_{i,h}(s, \mathbf{a})]$

Nash Equilibrium

Theorem

Players are in Nash equilibrium under π^ , with value of player i starting from state s at time h being $V_{i,h}(s)$. That is,*

$$V_{i,h}(s) = V_{i,h}^{\pi^*}(s) \geq V_{i,h}^{\pi'_i, \pi^{*-i}}(s), \quad \forall \pi'_i, s, h$$

Proof

ORCS 4529: Reinforcement Learning

└ Multi-Agent Reinforcement Learning (MARL)

└ Markov Games

└ Proof

Proof is by induction on h .

For $h = H$, player i 's "continuation" value starting from state s under π^* is given by

$$\mathbb{E}_{\mathbf{a} \sim \pi_H^*(s)}[R_i(s, \mathbf{a})] =: \mathbb{E}_{\mathbf{a} \sim \pi_H^*(s)}[Q_{i,H}(s, \mathbf{a})] =: V_{i,H}(s)$$

BY the algorithm, $\pi_{i,H}^* = p_i^*$ where p^* is a Nash equilibrium for the game matrix $Q_{i,H}(s, \cdot)$. Therefore, $V_{i,H}$ satisfies for all $p_i \in \Delta^{A_i}$:

$$\begin{aligned} V_{i,H}(s) &:= \mathbb{E}_{a_i \sim p_i^*, a_{-i} \sim p_{-i}^*}[Q_{i,H}(s, a_i, a_{-i})] &> \mathbb{E}_{a_i \sim p_i, a_{-i} \sim p_{-i}^*}[Q_{i,H}(s, a_i, a_{-i})] \\ &= \mathbb{E}_{a_i \sim p_i, a_{-i} \sim p_{-i}^*}[R_i(s, a_i, a_{-i})] \\ &= V_{i,H}^{p_i, p_{-i}^*}(s) \end{aligned}$$

Therefore, π_H^* is in Nash equilibrium and the value of player i is $V_{i,H}$.

ORCS 4529: Reinforcement Learning

- └ Multi-Agent Reinforcement Learning (MARL)
 - └ Markov Games
 - └ Proof

Now we prove for h by induction. Since the continuation value for player i from $h + 1$ is $V_{i,h+1}$, the value from h is

$$\begin{aligned}\mathbb{E}_{\mathbf{a} \sim \pi_{h:H}^*} [r_h + \gamma r_{h+1} + \dots + \gamma^{H-1} r_H | s_1 = s] &= \mathbb{E}_{\mathbf{a} \sim \pi_h^*} [R_i(s, \mathbf{a}) + \gamma \sum_{s'} P(s, \mathbf{a}, s') V_{i,h+1}(s')] \\ &= \mathbb{E}_{\mathbf{a} \sim \pi_h^*} [Q_{i,h}(s, \mathbf{a})] =: V_{i,h}(s)\end{aligned}$$

Also, since π_h^* is a Nash equilibrium of game matrix $Q_{i,h}$ and by induction since $V_{i,h+1} \geq V^{\pi_i, \pi_{-i}^*}$ for all π_i , we have

$$\begin{aligned}V_{i,h}(s) &= \mathbb{E}_{\mathbf{a} \sim \pi_h^*} [Q_{i,h}(s, \mathbf{a})] \\ &\geq_{NE} \mathbb{E}_{\mathbf{a} \sim \pi_{i,h}, \pi_{-i,h}^*} [Q_{i,h}(s, \mathbf{a})] \\ &= \mathbb{E}_{\mathbf{a} \sim \pi_{i,h}, \pi_{-i,h}^*} [R_i(s, \mathbf{a}) + \gamma \sum_{s'} P(s, \mathbf{a}, s') V_{i,h+1}(s')] \\ &\geq \mathbb{E}_{\mathbf{a} \sim \pi_{i,h}, \pi_{-i,h}^*} [R_i(s, \mathbf{a}) + \gamma \sum_{s'} P(s, \mathbf{a}, s') V_{i,h+1}^{\pi_i, \pi_{-i}^*}(s')] \\ &= V_{i,h}^{\pi_i, \pi_{-i}^*}(s)\end{aligned}$$

Infinite horizon discounted setting

Let $V_i^\pi(s)$ denote the value of player i under (stationary or non-stationary. Markovian or non-Markovian) policy profile $\pi = (\pi_1, \dots, \pi_m)$:

$$V_i^\pi(s) := \mathbb{E}\left[\sum_t \gamma^{t-1} R_i(s_t, \mathbf{a}_t) | s_1 = s; \pi\right]$$

Theorem (Fink 64, Takahashi 64)

Every infinite-horizon discounted, stochastic game with a finite number of states, actions, and players, has a Nash equilibrium in stationary, Markovian strategies. Specifically there exists a collection of stationary Markovian policies $\pi = (\pi_1, \dots, \pi_m)$ such that for every player i , any state s ,

$$V_i^\pi(s) \geq V_i^{\pi'_i, \pi^{-i}}(s)$$

However, computing this equilibrium is an open problem.

Proof

ORCS 4529: Reinforcement Learning

- └ Multi-Agent Reinforcement Learning (MARL)
 - └ Markov Games
 - └ Proof

Observation 1: For any i and stationary Markovian policy π , value vector V_i^π satisfies and is a unique solution of the equation

$$V_i^\pi = R_i^\pi + \gamma P^\pi V_i^\pi$$

where $R_i^\pi(s) = \mathbb{E}_{\mathbf{a} \sim \pi}[R_i(s, \mathbf{a})]$, $P^\pi(s, s') = \mathbb{E}_{\mathbf{a} \sim \pi}[P(s, \mathbf{a}, s')] = \sum_{\mathbf{a}} \pi(s, \mathbf{a}) P(s, \mathbf{a}, s')$. Uniqueness follows from the fact that the matrix $I - \gamma P^\pi$ is invertible for $\gamma < 1$.

Observation 2: V_i^π is continuous π (because $V_i^\pi = (I - \gamma P^\pi)^{-1} R_i^\pi$ is continuous in P^π which is continuous in π)

Observation 3: For every $a_i \in A_i$, let $Q_i^\pi(s, a_i) := \mathbb{E}_{a_{-i} \sim \pi_{-i}}[R_i(s, \mathbf{a}) + \gamma \sum_{s'} P(s, \mathbf{a}, s') V_i^\pi(s')]$. Define $\psi(\pi)$ as a mapping from Markovian stationary policy π to π' where for all $a \in A_i$,

$$\pi'_i(s, a) = \frac{\pi_i(s, a) + (Q_i^\pi(s, a) - V_i^\pi(s))^+}{1 + \sum_{a' \in A_i} (Q_i^\pi(s, a') - V_i^\pi(s))^+}$$

Then, ψ is a continuous mapping in π .

ORCS 4529: Reinforcement Learning

- └ Multi-Agent Reinforcement Learning (MARL)
 - └ Markov Games
 - └ Proof

Since ψ is a continuous mapping by Brouwer's fixed point theorem, there exists π^* such that

$$\pi^* = \psi(\pi^*)$$

At such a policy, for all $a \in A_i$,

$$Q_i^{\pi^*}(s, a) \leq V_i^{\pi^*}(s)$$

—— We show π^* is a Nash Equilibrium.

Define an MDP for player i , where for all $a_i \in A_i$,

$$\tilde{r}_i(s, a_i) = \mathbb{E}_{a_{-i} \sim \pi_{-i}^*}[R_i(s, a_i, a_{-i})]$$

$$\tilde{P}_i(s, a_i, s') = \mathbb{E}_{a_{-i} \sim \pi_{-i}^*}[P(s, a_i, a_{-i}, s')]$$

Let $\tilde{v}_i^{\pi_i}(s)$ be the value function in this MDP. Note that $\tilde{v}_i^{\pi_i^*} = V_i^{\pi^*}$. Observation 1 and 3 imply that

$$\tilde{v}_i^{\pi_i^*}(s) = \mathbb{E}_{a_i \sim \pi_i^*(s)}[\tilde{r}_i(s, a_i) + \gamma \tilde{P}_i(s, a_i, s') \tilde{v}_i^{\pi_i^*}(s)] = \max_{a_i}[\tilde{r}_i(s, a_i) + \gamma \tilde{P}_i(s, a_i, s') \tilde{v}_i^{\pi_i^*}(s)]$$

That is π_i^* is the optimal policy in this MDP. So, for all π_i , we have

$$V_i^{\pi^*}(s) = \tilde{v}_i^{\pi_i^*}(s) \geq \tilde{v}_i^{\pi_i}(s) = V_i^{\pi_i, \pi_{-i}^*}(s)$$

Zero-sum Markov Games (aka Stochastic Games)

A two-player zero-sum Markov Game is defined by tuple $(\mathcal{S}, A_1, A_2, P, R)$ [Shapley 1953]

- ▶ $\mathcal{S}$: set of states
- ▶ A_1, A_2 : set of actions for the max-player/the min-player.
- ▶ $P(s'|s, a_1, a_2)$: transition probability
- ▶ $R(s, a_1, a_2) \in (0, 1)$, the expected reward for the max player (loss for the min player).

Value function

Value of a policy profile $\pi = (\pi_1, \pi_2)$, given a (distribution over) starting state s :

$$V^{\pi_1, \pi_2}(s) = \mathbb{E}_{\pi_1, \pi_2} \left[\sum_{t=1}^{\infty} \gamma^{t-1} R(s_t, a_{1,t}, a_{2,t}) | s_1 = s \right]$$

Max player wants to maximize the value, min player wants to minimize the value.

ORCS 4529: Reinforcement Learning

└ Multi-Agent Reinforcement Learning (MARL)

└ Markov Games

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Max player wants to maximize the value, min player wants to minimize the value.

Here the expectation is over $a_{1,t} \sim \pi_1, a_{2,t} \sim \pi_2, s_{t+1} \sim P(\cdot | s_t, a_{1,t}, a_{2,t})$. Actions are sampled independently for the two players from their respective policies

Minimax theorem for Zero Sum Markov Games

Definition (Nash equilibrium of Markov Games)

Two policies π_1^*, π_2^* in Nash equilibrium if no player has an incentive to unilaterally deviate from their current policy. For all π_1, π_2 , and all s

$$V^{\pi_1, \pi_2^*} \leq V^{\pi_1^*, \pi_2^*} \leq V^{\pi_1^*, \pi_2}.$$

Theorem (Shapley's theorem for Markov games)

In two-player zero-sum Markov games, the minimax theorem holds:

$$\max_{\pi_1} \min_{\pi_2} V^{\pi_1, \pi_2} = \min_{\pi_2} \max_{\pi_1} V^{\pi_1, \pi_2}$$

*Any minimax point is a Nash equilibrium and vice-versa.
Furthermore, the minimax is achieved by Markovian policies.*

ORCS 4529: Reinforcement Learning

└ Multi-Agent Reinforcement Learning (MARL)

└ Markov Games

└ Minimax theorem for Zero Sum Markov Games

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above are vecors, inequalities are component-wise.

1. For any minimax point

$$V^{\pi_1^*, \pi_2^*} = \min_{\pi_2} \max_{\pi_1} V^{\pi_1, \pi_2} = \max_{\pi_1} V^{\pi_1, \pi_2^*} \geq V^{\pi_1, \pi_2^*} \text{ Similarly}$$

$$V^{\pi_1^*, \pi_2^*} = \max_{\pi_2} \min_{\pi_1} V^{\pi_1, \pi_2} = \min_{\pi_2} V^{\pi_1^*, \pi_2} \leq V^{\pi_1^*, \pi_2}. \text{ Therefore,}$$

π_1^*, π_2^* are in Nash Equilibrium.

2. For the other direction argue that if (π_1, π_2) is not a minimax point then either $V^{\pi_1, \pi_2} < V^{\pi_1', \pi_2}$ for some π_1' or $V^{\pi_1, \pi_2} > V^{\pi_1, \pi_2'}$ for some π_2' . So at least one of the player can improve their utility by defecting.

Minimax theorem

- ▶ Does not follow from Von Neumann's theorem - value function is not convex-concave.
- ▶ Proof via Dynamic Programming and Value iteration.

Proof

- └ Multi-Agent Reinforcement Learning (MARL)

- └ Markov Games

- └ Proof

$$V^{\pi_1, \pi_2} = \mathbb{E}_{\pi_1, \pi_2} \left[\sum_{t=1}^{\infty} \gamma^{t-1} R(s_t, a_{1,t}, a_{2,t} | s_1) \right]$$

Define new "L" operator:

$$[L_G v] = \max_{\alpha} \min_{\beta} \sum_{a,b} \alpha(a) \beta(b) (R(s, a, b) + \gamma \sum_{s'} P(s, a, b, s') v(s'))$$

Show that L_G operator is a gamma contraction. Then, by Banach's fixed point theorem there is a unique fixed point of $v = L_G v$, say V^* , and $L_G^k v \rightarrow V^*$ (by value iteration proof).

Define matrix M with entries $M(a, b) := (R(s, a, b) + \gamma \sum_{s'} P(s, a, b, s') V^*(s'))$. Then, by minimax theorem (Von Neumann).

$$[V^*](s) = [L_G V^*](s) = \max_{\alpha} \min_{\beta} \sum_{a,b} \alpha(a) \beta(b) M(a, b) = \min_{\beta} \max_{\alpha} \sum_{a,b} \alpha(a) \beta(b) M(a, b)$$

ORCS 4529: Reinforcement Learning

- └ Multi-Agent Reinforcement Learning (MARL)
 - └ Markov Games
 - └ Proof

So far we have proven there exists V^* such that

$$V^* = \max_{\alpha} \min_{\beta} \sum_{a,b} \alpha_s(a) \beta_s(b) (R(s, a, b) + \gamma \sum_{s'} P(s, a, b, s') V^*(s')) = \min_{\beta} \max_{\alpha} \sum_{a,b} \alpha(a) \beta(b) (R(s, a, b) + \gamma \sum_{s'} P(s, a, b, s') V^*(s'))$$

For any α, β , consider the corresponding policies. Then, we show that value of both players under these policies is: (i.e., simple proof by dynamic programming)

$$V^{\alpha, \beta} = \sum_{a,b} \alpha_s(a) \beta_s(b) (R(s, a, b) + \gamma \sum_{s'} P(s, a, b, s') V^{\alpha, \beta}(s'))$$

Also, $V^{\alpha, \beta}$ is unique solution to this equation. Therefore from above result, Let α^*, β^* achieve the minimax, then

$$V^{\alpha^*, \beta^*} = V^*$$

Therefore,

$$V^{\alpha^*, \beta^*} = V^*$$

Computing Nash Equilibrium

A milder goal: ϵ -Nash equilibrium

Definition

An ϵ -Nash equilibrium is a policy profile $\hat{\pi} = (\hat{\pi}_1, \hat{\pi}_2)$ such that

$$\max_{\pi_1} V^{\pi_1, \hat{\pi}_2} - \min_{\pi_2} V^{\hat{\pi}_1, \pi_2} \leq \epsilon$$

Every player can gain at most ϵ by defecting

ORCS 4529: Reinforcement Learning

- └ Multi-Agent Reinforcement Learning (MARL)
 - └ Markov Games
 - └ Computing Nash Equilibrium

Computing Nash Equilibrium

A milder goal: ϵ -Nash equilibrium

Definition

An ϵ -Nash equilibrium is a policy profile $\pi = (\pi_1, \pi_2)$ such that

$$\max_{\pi_1} V^{\pi_1, \pi_2} - \min_{\pi_2} V^{\pi_1, \pi_2} \leq \epsilon$$

Every player can gain at most ϵ by defecting

For any approximate Nash, my defecting max player can achieve atmost :

$$\max_{\pi_1} V^{\pi_1, \hat{\pi}_2} \leq \min_{\pi_2} V^{\hat{\pi}_1, \pi_2} + \epsilon \leq V^{\hat{\pi}_1, \hat{\pi}_2} + \epsilon.$$

Similarly for the min player.

Computing (approximate) Nash equilibrium in Markov Games

Best response computation:

Given π_2 best response of player 1,

$$\arg \max_{\pi_1} V^{\pi_1, \pi_2}$$

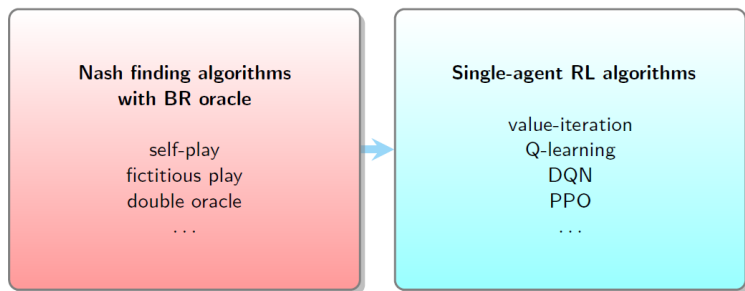
For a fixed π_2 , this is a single agent RL problem!

Algorithms for normal form games

- ▶ Best response dynamics aka self-play
- ▶ Fictitious play
- ▶ Double oracle
- ▶ weighted majority (aka multiplicative weight update)
- ▶ Gradient descent or Mirror descent

General Recipe

Key insight: given a fixed opponent, computing the best response (BR) is a single-agent RL problem.



commonly used in practice

Self play

Self-play

for $k = 1, \dots, K$,

$$\pi_1^{k+1} = BR(\pi_2^k).$$

$$\pi_2^{k+1} = BR(\pi_1^{k+1}).$$

π_i^k : the policy of the i^{th} player at
the k^{th} iteration

Does not converge even for rock-paper-scissors!

Fictitious play

In every iteration, computes the best response to the average policy (randomized) of the opponent.

Fictitious play [Brown, 1949]

for $k = 1, \dots, K$,

$$\pi_1^{k+1} = BR[(1/k) \cdot (\pi_2^1 + \dots + \pi_2^k)].$$

$$\pi_2^{k+1} = BR[(1/(k+1)) \cdot (\pi_1^1 + \dots + \pi_1^{k+1})].$$

How will this behave in rock-paper-scissors?

Asymptotic convergence

Theorem (Robinson 1951)

Fictitious play converges to Nash equilibrium.

- ▶ Inspecting the proof of [Robinson 1951], it requires $(1/\epsilon^{\Omega(A)})$ iterations to converge to ϵ -Nash equilibrium for a normal-form game with A actions.
- ▶ Karlin conjectured in 1959 that this rate can be improved to $O(1/\epsilon^2)$.
- ▶ Daskalakis and Pan [2014] refute the conjecture and prove that $(1/\epsilon^{\Omega(A)})$ iterations are required in the worst case.

Double oracle

Represents a class of approaches that use more informed weights than fictitious play.

$$M_k = \begin{matrix} & \dots & \pi_2^j & \dots \\ \vdots & & \vdots & \\ \pi_1^i & \left(\begin{array}{ccc} \dots & V^{\pi_1^i, \pi_2^j} & \dots \\ \vdots & & \vdots \end{array} \right) & \\ \vdots & & \vdots & \end{matrix}$$

Double Oracle

for $k = 1, \dots, K$,

$\mathbf{p}, \mathbf{q} \leftarrow$ a Nash equilibrium of M_k .

$$\pi_1^{k+1} = BR[\sum_{i=1}^k p_i \pi_1^i].$$

$$\pi_2^{k+1} = BR[\sum_{j=1}^k q_j \pi_2^j].$$

Can use some efficient algorithmic approach that may not be based on BR (like weighted majority) for solving the matrix game.

Double oracle convergence

Convergence of double oracle [McMahan 2003]

Double oracle algorithm finds Nash equilibrium of a normal-form game with A actions in $\mathcal{O}(A)$ iterations.

Is that good enough?

- ▶ Here A is the size of action or strategy space in normal form games.
- ▶ Directly converting a MG into a norm-form game gives $A = (1/\epsilon)^{HSA}$ – the size of policy space.

Drawbacks of the direct reduction approach

- ▶ Algorithms are designed based on black-box usage of single-agent RL, which does not exploit the detailed structure of MGs.
- ▶ Converting a MG into a norm-form game gives a number of actions $(1/\epsilon)^{HSA}$.
- ▶ Finding BR is NOT always a easy single-agent RL problem:
 - ▶ Even though Nash equilibrium is always achieved by a Markov policy, best response to a given policy may not.
 - ▶ When the min-player deploys a fixed non-Markovian policy, the game is NOT an MDP from the perspective of the max-player.
 - ▶ Existing single-agent RL results do not apply.

ORCS 4529: Reinforcement Learning

└ Multi-Agent Reinforcement Learning (MARL)

└ Markov Games

└ Drawbacks of the direct reduction approach

Drawbacks of the direct reduction approach

- ▶ Algorithms are designed based on black-box usage of single-agent RL, which does not exploit the detailed structure of MGs.
- ▶ Converting a MG into a norm-form game gives a number of actions $(1/v)^{NMA}$.
- ▶ Finding BR is NOT always a easy single-agent RL problem:
 - ▶ Even though Nash equilibrium is always achieved by a Markov policy, best response to a given policy may not.
 - ▶ When the min-player deploys a fixed non-Markovian policy, the game is NOT an MDP from the perspective of the max-player.
 - ▶ Existing single-agent RL results do not apply.

this means $O(A)$ is not a good complexity. And algorithms like weighted majority cannot be applied

Bellman Optimality equations

Nash equilibrium of MG is a pair of Markovian policies (possibly randomized), satisfying the Bellman equation.

Finite horizon

Dynamic Programming

- ▶ Define $V_k^*(s)$ as the (discounted) value of playing minimax Markovian (non-stationary) policy starting from state s and horizon k : i.e.,

$$V_k^*(s) := \max_{\pi_1} \min_{\pi_2} \mathbb{E}_{\pi_1, \pi_2} \left[\sum_{t=1}^k \gamma^{t-1} r_t \mid s_1 = s \right]$$

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- ▶ $V_k^*(s) = V_k^{\pi_1^*, \pi_2^*}(s)$ is the value of the k -stage “subgame” starting from state s .

Finite horizon

Dynamic Programming

For a (sub-)game starting from state s , value of game is given by

$$V_k^*(s) = \max_{p \in \Delta(A_1)} \min_{q \in \Delta(A_2)} \mathbb{E}_{a_1 \sim p, a_2 \sim q} [R(s, a_1, a_2) + \gamma P(s, a_1, a_2) V_{k-1}^*]$$

where $V_0^* = 0$.

- ▶ Using minimax theorem, the order of max and min in the above can be reversed.
- ▶ The $\arg \max p$ and $\arg \min q$ above form equilibrium (non-stationary) policies.

DP for Finite Horizon H

In terms of Q-values. Define

$$Q_k^*(s, a_1, a_2) = R(s, a_1, a_2) + \gamma P(s, a_1, a_2) V_{k-1}^*$$

Then,

$$\begin{aligned} V_k^*(s) &= \max_{p \in \Delta(A_1)} \min_{q \in \Delta(A_2)} \sum_{a_1, a_2} p(a_1) Q_k^*(s, a_1, a_2) q(a_2) \\ &= \max_{p \in \Delta(A_1)} \min_{q \in \Delta(A_2)} p^T Q_k^*(s) q \\ &= \text{Nash-value of two player matrix Game } Q_k^*(s) \end{aligned}$$

where $Q_k^*(s)$ is the matrix with entries $Q_k^*(s, a_1, a_2)$.

Proof

Finite horizon setting

Ininite horizon discounted setting

Bellman equations

- ▶ Define $V^*(s)$ as the value of playing minimax Markovian policy starting from state s : i.e.,

$$V^*(s) := \max_{\pi_1} \min_{\pi_2} V^{\pi_1, \pi_2}(s) = \max_{\pi_1} \min_{\pi_2} \mathbb{E}_{\pi_1, \pi_2} \left[\sum_{t=1}^{\infty} \gamma^{t-1} r_t \mid s_1 = s \right]$$

$V^*(s)$ gives value of the “subgame” starting from state s .

Infinite horizon discounted setting

Bellman equations

$$V^*(s) = \max_{p \in \Delta(A_1)} \min_{q \in \Delta(A_2)} \mathbb{E}_{a_1 \sim p, a_2 \sim q} [R(s, a_1, a_2) + \gamma P(s, a_1, a_2) V^*]$$

- ▶ The arg max p and arg min q above form equilibrium (non-stationary) policies with $\pi_1^*(s) = p$, $\pi_2^*(s) = q$.
- ▶ Using minimax theorem, the order of max and min in the above can be reversed so that

$$V^*(s) = \min_{\pi_2} \max_{\pi_1} V^{\pi_1, \pi_2}(s) = \max_{\pi_1} \min_{\pi_2} V^{\pi_1, \pi_2}(s)$$

Bellman equations

In terms of Q-values

Define

$$Q^*(s, a_1, a_2) = R(s, a_1, a_2) + \gamma P(s, a_1, a_2) V^*$$

Then,

$$\begin{aligned} V^*(s) &= \max_{p \in \Delta(A_1)} \min_{q \in \Delta(A_2)} \sum_{a_1, a_2} p(a_1) Q^*(s, a_1, a_2) q(a_2) \\ &= \max_{p \in \Delta(A_1)} \min_{q \in \Delta(A_2)} p^T Q^*(s) q \\ &= \text{Nash-value of two player matrix Game } Q^*(s) \end{aligned}$$

where $Q^*(s)$ is the matrix of $Q^*(s, a_1, a_2)$.

Will inspire Nash Value Iteration, Nash Q-learning etc.

Proof of Bellman equations

inifinte horizon discounted setting

1. There exists a V^*, Q^* satisfying the Bellman equation.
2. For any V^* satisfying the Bellman equation, $V^*(s)$ gives the value of the “subgame” starting from state s , i.e.,

$$V^*(s) = \max_{\pi_1} \min_{\pi_2} V^{\pi_1, \pi_2}(s) = \min_{\pi_2} \max_{\pi_1} V^{\pi_1, \pi_2}(s)$$

3. Equilibrium policy π_1^*, π_2^* is given by

$$(\pi_1^*(s), \pi_2^*(s)) := \arg \max_{p \in \Delta(A_1)} \min_{q \in \Delta(A_2)} p^T Q^*(s) q$$

ORCS 4529: Reinforcement Learning

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 - Proof of Bellman equations

Proof of Bellman equations

infinite horizon discounted setting

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In fact, the proof will be constructive showing that the k -stage value function will converge to V^* as $k \rightarrow \infty$

Nash Q-learning [Hu and Wellman 2003]

Initialize Q .

For $t = 1, 2, \dots$

- ▶ Observe state s_t .
- ▶ Take action $a_t = (a_{1,t}, a_{2,t})$.
- ▶ Observe reward r_t and state s_{t+1} .
- ▶ Q-learning update:

$$Q_{t+1}(s_t, a_t) \leftarrow (1 - \alpha_t)Q_t(s_t, a_t) + \alpha_t(r_t + \gamma \text{Nash}(Q_t(s_{t+1}, \cdot, \cdot)))$$

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where $\text{Nash}(Q_t(s_{t+1}, \cdot, \cdot))$ is the Nash-value of the two-player zero-sum game

$$\text{Nash}(Q_t(s_{t+1}, \cdot, \cdot)) = \max_{p \in \Delta(A_1)} \min_{q \in \Delta(A_2)} p^T Q_t(s_{t+1}, \cdot, \cdot) q$$

Nash Q-learning [Hu and Wellman 2003]

Initialize Q .

For $t = 1, 2, \dots$

- ▶ Observe state s_t .
- ▶ Take action $a_t = (a_{1,t}, a_{2,t})$.

Greedy action:

$$(a_{1,t}, a_{2,t}) = \text{Nash}(Q_t(s_t, \cdot, \cdot))$$

- ▶ Observe reward r_t and state s_{t+1} .
- ▶ Q-learning update:

$$Q_{t+1}(s_t, a_t) \leftarrow (1 - \alpha_t)Q_t(s_t, a_t) + \alpha_t(r_t + \gamma \text{Nash}(Q_t(s_{t+1}, \cdot, \cdot)))$$

Convergence

(Szepesvari and Littman (1999), Corollary 5)

Theorem

Given

- ▶ $\sum_{t=1}^{\infty} \alpha_t = \infty$, $\sum_{t=1}^{\infty} \alpha_t^2 < \infty$ *and*
- ▶ *Every state s and actions $a_1 \in A_1, a_2 \in A_2$ are visited infinitely often.*

Then, we have that the Q-learning converges, $Q_t \rightarrow Q^$.*

ORCS 4529: Reinforcement Learning

- └ Multi-Agent Reinforcement Learning (MARL)
 - └ Markov Games
 - └ Convergence

Convergence
(Sutton and Littman (1998), Corollary 5)

Theorem
Given

- ▶ $\sum_{t=1}^{\infty} \alpha_t = \infty$, $\sum_{t=1}^{\infty} \alpha_t^2 < \infty$ and
- ▶ Every state s and actions $a_1 \in A_1, a_2 \in A_2$ are visited infinitely often.

Then, we have that the Q-learning converges, $Q_t \rightarrow Q^*$.

Note that convergence does not depend on which actions are played, as long as all states and actions are visited infinitely often. However in order to ensure that in an efficient manner, we need an ϵ -greedy like action. More recent papers like Sidford et al 2019 analyze sample complexity of Nash-Q-learning type algorithms